

Lab Report-1

Experiment 1: Familiarization with Fundamental Logic Gates

Objective:

- To get familiarized with fundamental logic gates and demonstrate the input-output relationship of 2-input AND (IC-7408), OR (IC-7432) and NOT/Inverter (IC-7404) gates by constructing their truth tables.
- To get familiar with other logic gates like NAND (IC-7400), NOR (IC-7402), XOR (IC-7486) and XNOR (IC-4077)

Required Components:

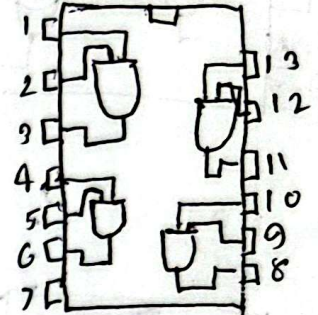
- | | | |
|---------------|---------------|---------------|
| 1. IC 7408 x1 | 4. IC 7400 x1 | 7. IC 4077 x1 |
| 2. IC 7432 x1 | 5. IC 7402 x1 | |
| 3. IC 7404 x1 | 6. IC 7486 x1 | |

Discussions :

In the experiment, we drew and tested the circuits using bread, board and trainer board, we had set up every gc's in the breadboard, then we tested the outputs with the given truth table. Some of the gc's have a bit different diagram, such as, the inputs and outputs of gc 7402 and 7432 are different. Thus, we had to carefully place the input and output wires on the board, In the end, it was a fun experiment.

Experimental Setup & Truth Table AND

X	Y	Q (output)
0	0	0
0	1	0
1	0	0
1	1	1

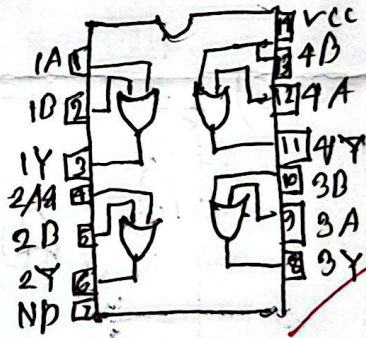


SP.

OR

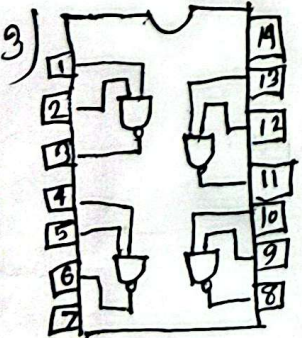
2)

x	y	output
0	0	0
1	0	1
0	1	1
1	1	1



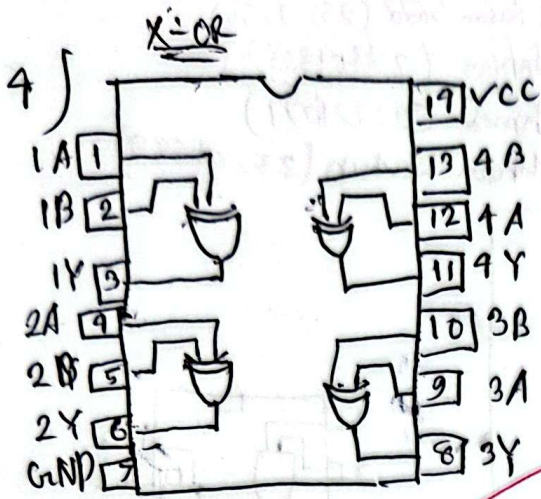
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NAND



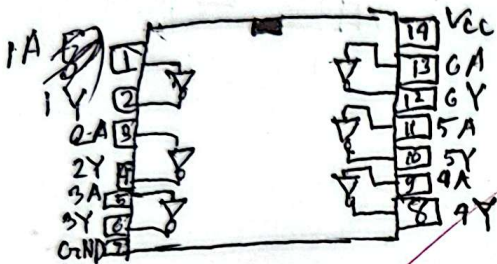
x	y	output
0	0	1
1	0	1
0	1	1
1	1	0

SP.



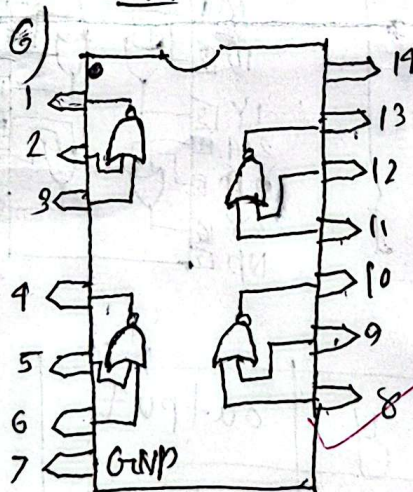
x	y	output
0	0	0
1	0	1
0	1	1
1	1	0

5) NOT



Input	output
A	
0	1
1	0

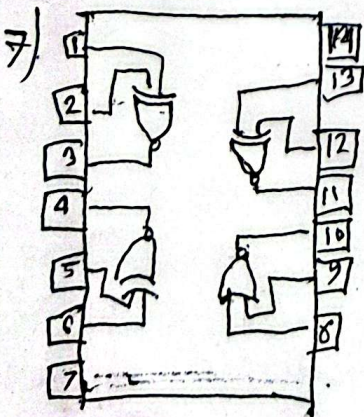
NOR



x	y	output
0	0	1
1	0	0
0	1	0
1	1	0

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X-NOR



x	y	output
0	0	1
1	0	0
0	1	0
1	1	1

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Experiment 2: Universal Gates, Applications of Boolean Algebra

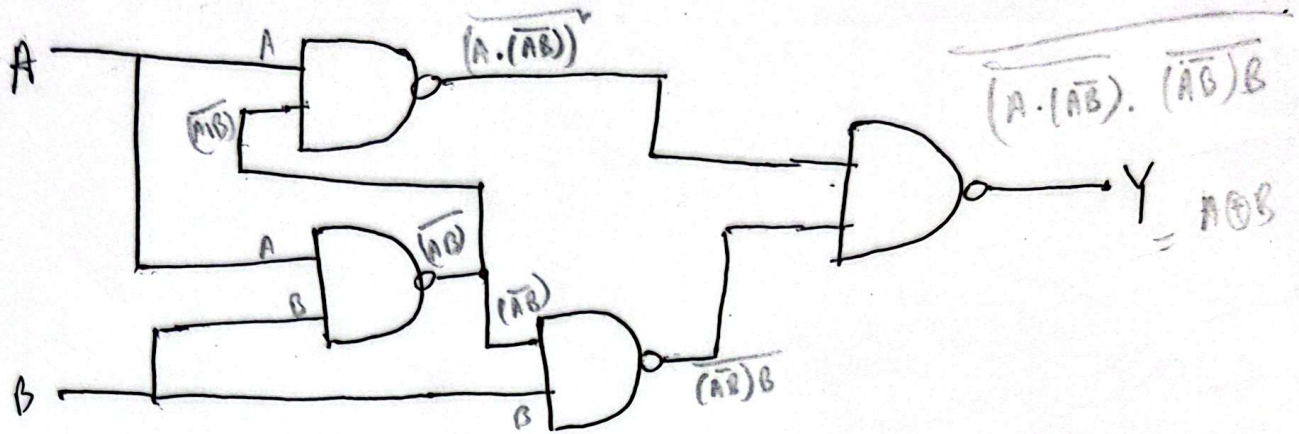
Objective:

- To investigate the rules of Boolean Algebra
- To gain experience working with practical circuits
- To simplify a complex function using Boolean algebra

Required Components:

1. IC 7400 x1
2. IC 7402 x2

Experimental setup, Truth table, Boolean equations, and Discussion



$$((A.(AB)')'.(AB).(A'))'$$

$$= (A.(AB)') + (AB.(A'))$$

$$= (A.(A'+B')) + (B.(A'+B'))$$

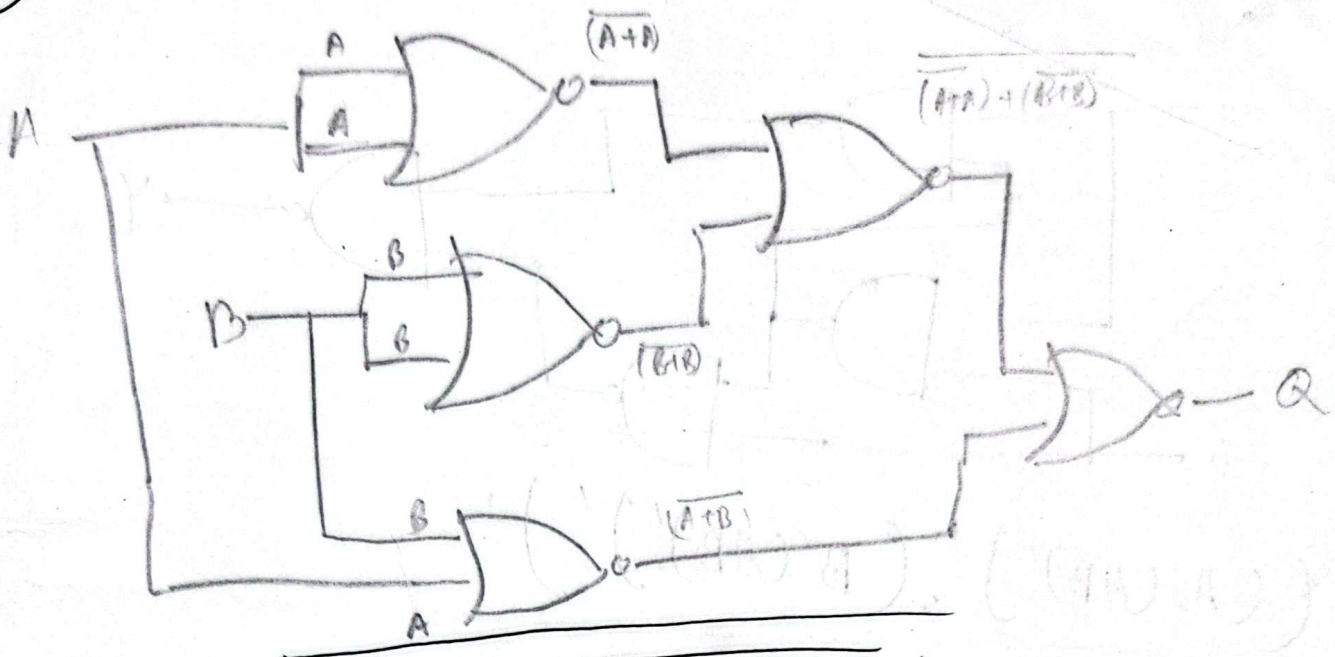
$$= AA' + AB' + A'B + BB'$$

$$= AB' + A'B$$

$$= A \oplus B \quad [\text{XOR Gate}] //$$

input		output
A	B	
0	0	0
1	0	1
0	1	1
1	1	0

(2)



$$F = \left(\overline{A+B} + \overline{A \cdot A + B \cdot B} \right)$$

$$= \left(\overline{A+B} + \overline{A+B} \right)$$

$$= \left(\overline{A+B} + \overline{A \cdot B} \right)$$

$$= \left(\overline{A+B} + A \cdot B \right)$$

$$= \left(\overline{A} \cdot \overline{B} + A \cdot B \right)$$

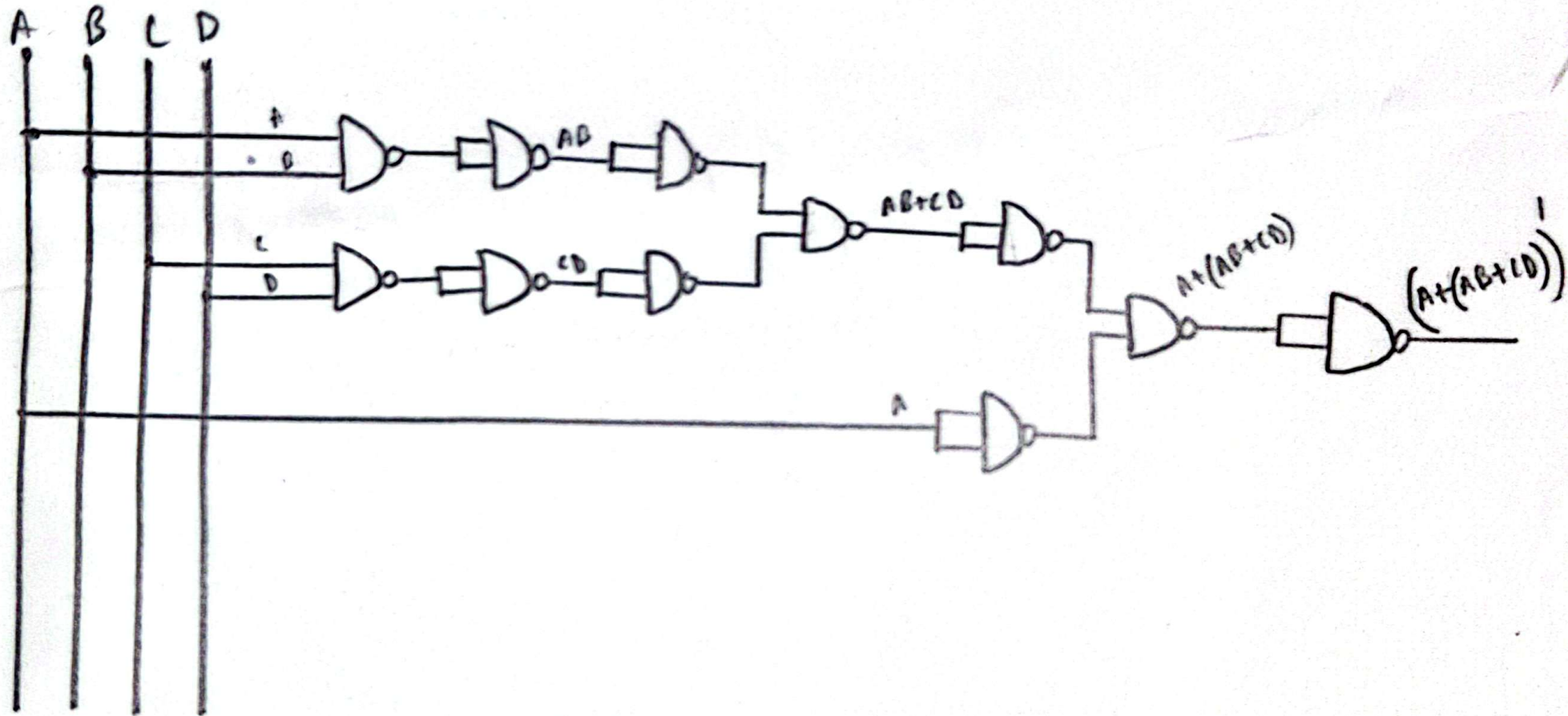
$$= \overline{A} \cdot \overline{B} + A \cdot B$$

$$= (A+B) \cdot (\overline{A} + \overline{B})$$

$$= A \cdot \overline{A} + A \overline{B} + \overline{A} B + B \overline{B} = A \overline{B} + \overline{A} B$$

$$= A \oplus B \quad \text{[XOR Gate]}$$

d) $(A + (AB + CD))'$



Experiment no: 3

Experiment Name: Parity Bit Checker and Generator

Objective: Using XOR gates (IC-7486) design and implement an even parity checker and generator.

Required Components:

- i) IC - 7486 x 1
- ii) Trainer Board
- iii) Bread Board
- iv) Jumper wire

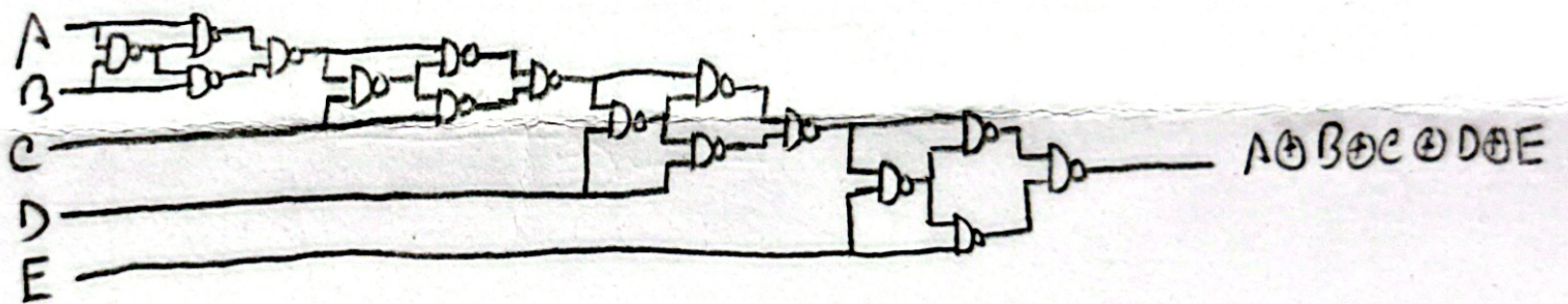
Discussions:

(a) 3-bit parity generator using NOR

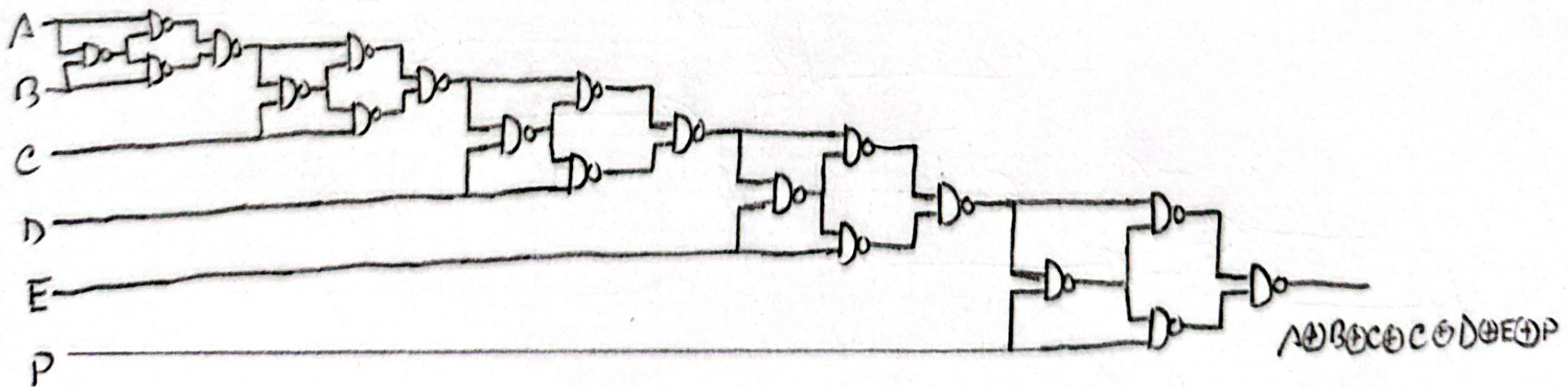
Discussion: (a): Answered in the question part.

(b) 5-bit parity checker and generator using NAND gate.

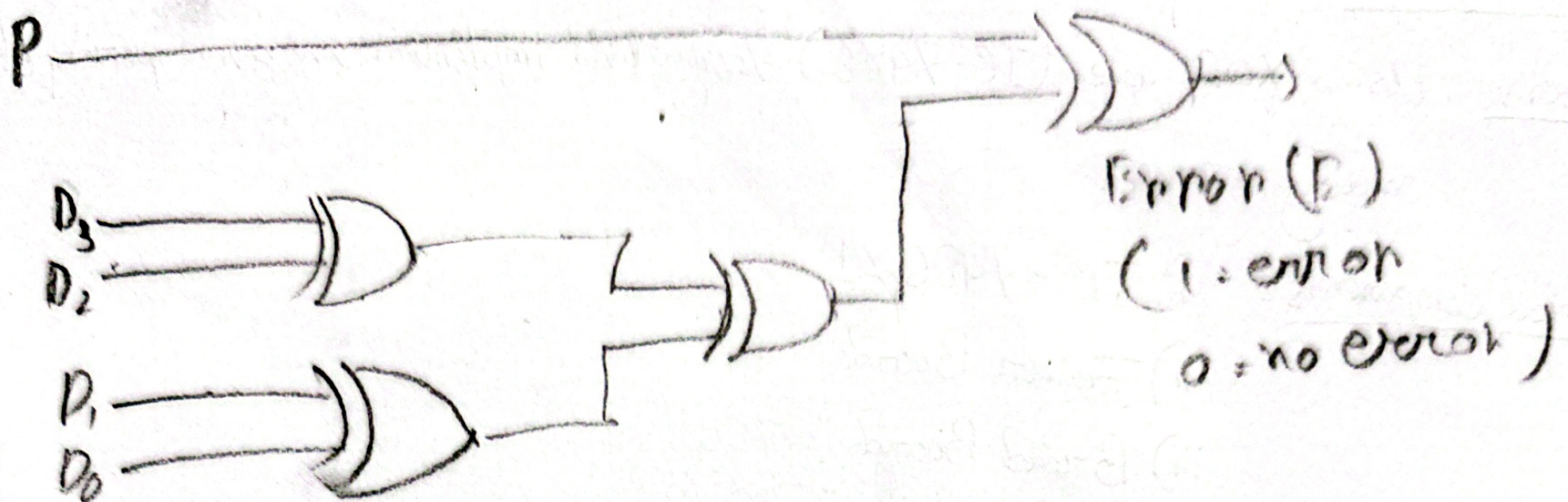
i) Generator:



ii) checker:



Building an even parity checker using XOR gates for 4-bit Data:



Truth table for parity Bit Generator

	Data				Parity
	D ₃	D ₂	D ₁	D ₀	
a	1	0	0	1	0
b	0	0	0	1	1
c	1	1	1	1	0
d	0	0	0	0	0

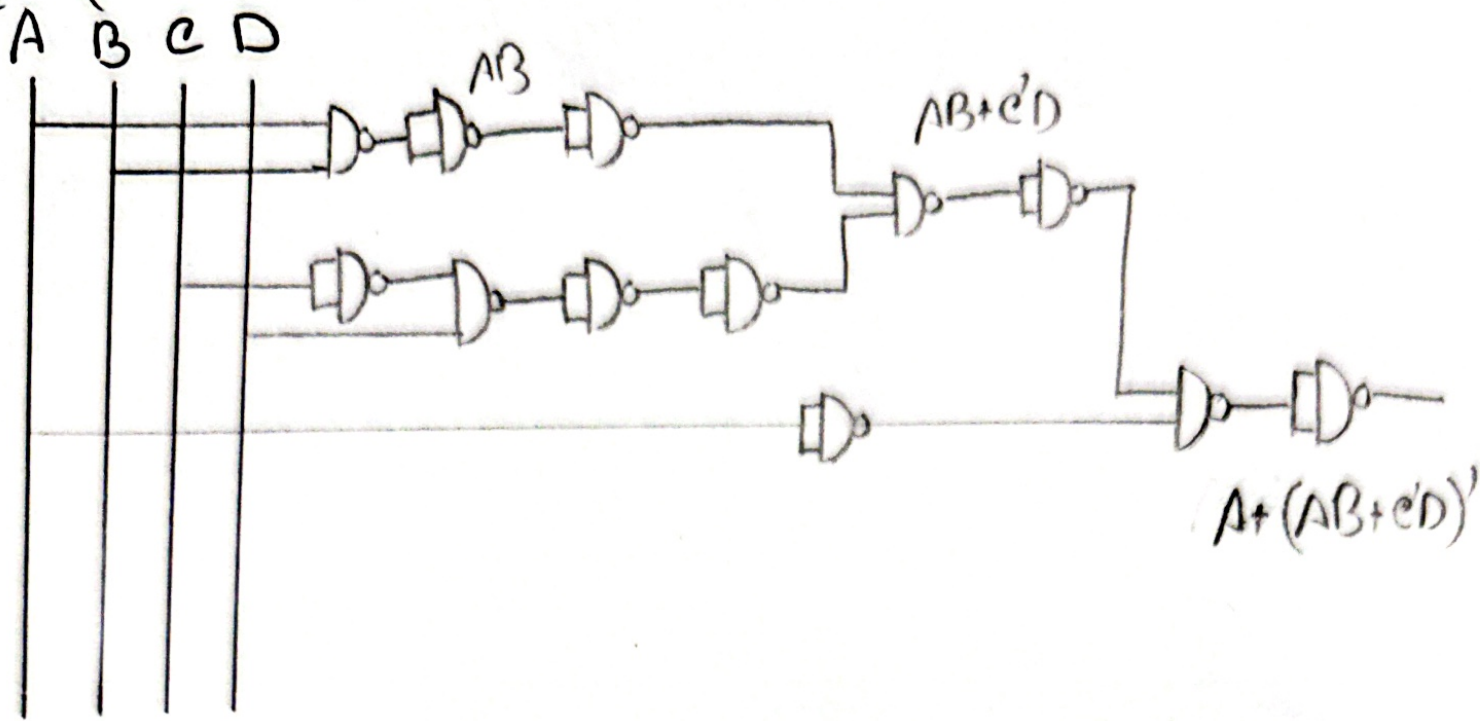
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Truth Table for Parity Bit Checker

							Error
	Parity	D ₃	D ₂	D ₁	D ₀		
a	1	1	0	0	1		1
b	0	0	0	0	1		1
c	0	1	1	1	1		0
d	1	0	0	0	0		1

Ans. the following question:

1) $(A + (AB + C'D))'$



2) Circuit Diagram of 3-bit parity generator using NOR gate:

